

Design Research Report

Integration of AI Models

Specific AI Models

1. **Facial Expression Emotion Detection:**
 - **Model:** Hugging Face dima806/facial_emotions_image_detection.
 - **Role:** Processes video frames from the user's camera to detect facial emotions in real-time.
2. **Speech-to-Text Transcription:**
 - **Model:** Hugging Face openai/whisper-small.
 - **Role:** Transcribes audio inputs from the user's microphone into text for further sentiment analysis.
3. **Text Sentiment Analysis:**
 - **Model:** Hugging Face cardiffnlp/twitter-roberta-base-sentiment-latest.
 - **Role:** Analyzes the sentiment of transcribed text to determine emotional tones (Positive, Neutral, Negative).
4. **Emotion-Driven Text Generation:**
 - **Model:** OpenAI GPT-4.
 - **Role:** Combines the results from facial emotion and text sentiment analysis to generate supportive and encouraging text responses.

Workflow Integration

1. **Real-Time Inputs:**
 - Video and audio inputs are captured dynamically via the browser.
2. **Model Interaction:**
 - Facial emotion detection and speech-to-text transcription are processed in parallel.
 - Text sentiment analysis combines with facial sentiment to provide contextual input for GPT-4.
3. **Response Generation:**
 - GPT-4 generates an emotionally supportive message based on the combined analysis.

Rationale for Technical Decisions

AI Model Selection

- **Hugging Face:** Pretrained models like dima806/facial_emotions_image_detection and cardiffnlp/twitter-roberta-base-sentiment-latest offer high accuracy for emotion and sentiment detection.
- **OpenAI Whisper:** Known for its robust transcription capabilities, crucial for converting audio to text.
- **OpenAI GPT-4:** Provides context-aware, empathetic responses, aligning with the project's objective to enhance emotional support.

Technical Stack

- **Frontend:** p5.js and HTML for seamless video and audio capture and dynamic visual interactions.
- **Backend:** Flask for managing API calls and integrating AI models.
- **Integration Challenges:** Focused on ensuring smooth communication between the frontend and backend, as well as seamless data flow between the AI models.

Results from the Preliminary Study

Findings

1. **User Preferences:**
 - Users predominantly seek emotional support rather than actionable suggestions, especially in gaming contexts.
 - Positive reinforcement and empathetic messages were valued over strategic advice.
2. **Challenges Identified:**
 - Real-time feedback required a robust pipeline for integrating multiple AI models.
 - Ensuring the latency of responses did not disrupt user interaction.

Impact on Final Design

- Shifted the focus toward enhancing the emotional support functionality.
- Simplified predictive features to emphasize empathetic engagement.

Challenges and Solutions

Major Challenges

1. **Frontend-Backend Communication:**
 - **Problem:** Synchronizing data between video/audio inputs in the frontend and model processing in the backend.
 - **Solution:** Implemented RESTful APIs in Flask to handle asynchronous data exchange.
2. **Inter-Model Communication:**
 - **Problem:** Ensuring data compatibility between facial emotion detection, speech-to-text transcription, and text sentiment analysis.
 - **Solution:** Used standardized formats (JSON) to streamline data flow across models.
3. **Latency Issues:**
 - **Problem:** Delays in real-time processing affected user experience.
 - **Solution:** Optimized model loading and used multithreading to parallelize tasks.

Inspiration

1. **Personal Experience:**
 - Inspired by the emotional highs and lows encountered in FPS games like *Counter-Strike*. Observing players' need for encouragement after losses

or mistakes informed the focus on emotional support.

2. External Project Influence:

- *Dead in Iraq: It's No Game*: This online protest and memorial within *America's Army* inspired the inclusion of a reflective, empathetic layer to the application. The emotional engagement demonstrated by this project emphasized the importance of meaningful interactions in a digital medium.